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Original Article

Tablet personal computer distraction during intravenous placement for young children in the pediatric emergency department: A pilot study

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Abstract ***Background***: Intravenous (IV) placement is a common procedure experienced by children visiting the pediatric emergency department (PED). However, uncontrolled anxiety and pain cause children to interfere with the procedure. In this pilot study, we sought to evaluate the effectiveness of tablet personal computers as a distraction method during IV placement.

Methods: This is a single-center pilot study conducted at a tertiary teaching hospital. Children visiting the PED were eligible if they were aged 3–5 years and required IV placement during the PED visit. After written consent was obtained from the guardian, the child was randomly assigned to a control group or an intervention group. For the intervention group, an animated video was played via tablet PC during IV placement. For both groups, children's anxiety, heart rate, and pain scale scores (the Face, Legs, Activity, Cry, Consolability and Evaluation Enfant Douleur) and guardian satisfaction were recorded.

Results: 22 children were eligible for the final analysis. There was no significant difference in the pain scale scores between the two groups, with the exception of the degree of pain relief after the procedure measured using Evaluation Enfant Douleur (intervention group: 6.0, interquartile range (IQR): 4.2–6.8, and control group; 3.0, IQR: 2.0–3.8, P = 0.011) and Face, Legs, Activity, Cry, Consolability (intervention group: 4.0, IQR: 4.0–4.2 and control group; 3.0, IQR: 1.5–3.5, P = 0.043).

Conclusion: In this pilot study, distraction using tablet personal computers may have reduced children's distress during the recovery phase after venipuncture. Further study with a larger sample size and different methods of distraction is essential.

Key words emergency room, needle stick, pain, pediatrics, tablet computer.

Intravenous (IV) placement is one of the most common invasive procedures experienced by children visiting a pediatric emergency department (PED).[1,2] It is difficult for even skilled physicians to perform venipuncture in young children who do not cooperate. Uncontrolled anxiety and pain related to needle procedures cause children to move more, which increases the chance of failure.[3] Previous studies have shown that various types of distraction methods have meaningful effects, reducing children's distress during a needle procedure.[1,4–6] In particular, non-pharmacologic distraction methods using objects familiar to children, such as toys, dolls, and electronic devices, are widely used in PEDs.[7–9]

Recently, the use of tablet personal computers (PCs) has become increasingly common among children. Guardians often participate actively in distracting their children from a needle procedure in the PED by showing their children's favorite videos using a tablet PC. The distraction method using a tablet PC seems to be able to provide visual and auditory stimulation effectively that diverts children's attention from the distress of the procedure and directly blocks the needle procedure scene itself. The tablet PC may become one of the most commonly used distraction tools in the near future because it is easily portable and allows children to choose personalized content.

Although many studies have focused on the use of tablet PCs for distraction, the strength of the evidence is not sufficient despite expectations of positive effects.[10–12] In addition, few studies have been performed using tablet PCs in PEDs, the most challenging environment for young children to experience needle procedures. Considering previous studies, improvements in the study protocol and standardization of outcome measurement were required to demonstrate the efficacy of tablet PC distraction appropriately. In the current study, we aimed to assess the feasibility and acceptability of distraction methods using a tablet PC during IV placement in PEDs and

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to identify whether they contribute to significant distress reduction during IV placement.

Methods

Study design and setting

This single-center pilot study was conducted at our children's hospital PED in Seoul, South Korea, from September 2017 to December 2017. Our hospital is an urban, tertiary academic hospital with a 315-bed capacity, and its emergency department (ED) receives more than 20 000 children per year. This study was approved by the institutional review board (IRB no. 1706-195-865) and was registered in the Korean Clinical Trial Registry (KCT no. 0006195).

Participants

Children visiting the PED were eligible if they were aged 3–5 years and required IV placement during the PED visit. Children with known vision or hearing impairment, developmental disabilities or facial anomalies, or chronic medical conditions requiring frequent IV placements were excluded from the study. Children were also excluded if they needed urgent IV placement (e.g., due to an unstable haemodynamic condition or an altered mental status) or if their guardians had insufficient Korean language ability to understand the study protocol. One research team member determined eligibility by screening patients visiting the PED during working hours (09:00 AM–4:00 PM, weekdays). If a child was eligible, the team member explained the aim and protocol of the study to the guardian and obtained written consent.

Randomization and study protocol

After written consent was obtained from the guardian, children were randomized to either the intervention group or the control group according to a computer-generated randomization sequence using the randomization.com website (http://www.randomization.com). The study protocol after group assignment is depicted in Figure 1. After the child entered the blood collection room, a physician began the video recording immediately. Within 5 min of the start of IV placement, the research personnel measured the child's baseline heart rate. For those children assigned to the intervention group, a physician turned on the animation using a tablet PC within 1 min after starting the video recording. The emergency medical technician (EMT) began IV placement. Approximately 15 s after starting the IV placement, the second heart rate was recorded. Finally, 1 min after completing the IV placement, the last heart rate was recorded. The animation and video recording ended 1 and 2 min after the last data were collected, respectively. After the child left the room, the guardian received and completed a questionnaire about the needle procedure. For the control group, all the procedures were conducted as in the intervention group but without showing the animation. The recorded video was collected and given to two board-certified emergency medicine physicians who were blinded to the study protocol. Those two physicians assessed the child's pain at the same time that each heart rate was recorded. Throughout the study period, the occurrence of any serious adverse events and side effects was documented.

Tablet PC and animation

The intervention group watched an animated show using a tablet PC (iPad Pro, Apple, Inc.), shared for research purposes only in the PED. The size of the tablet PC was 305.7 mm wide by 220.6 mm tall. The scene of the procedure was therefore sufficiently covered from the child's field of view. The tablet PC was fixed at a distance of approximately 30 cm in front of the child's face, and the child could watch the screen while lying down during the procedure without using his or her hands.

To focus on the distraction effect of the tablet PC itself and minimize confounding by content, all children watched the same animated video via the tablet PC. The content used in this study was a famous Korean animated show entitled "Baby Shark". We used a free video available from the official Pinkfong channel on the online video sharing platform YouTube (https://youtu.be/761ae_KDg_Q). We received approval from the production company for the use of the episode for research purposes prior to the study period.

Variables

Basic demographic variables such as age, gender, reason for the child's PED visit, whether the child had previous venipuncture experience, baseline anxiety assessed using the modified Yale Preoperative Anxiety Scale (mYPAS)[13] and heart rate were collected. Guardians' characteristics were also collected, such as age, relationship with the child, number of children, and previous experience of observing venipuncture in a child. The heart rate and observed pain were assessed at three time points during the IV placement: pre-IV placement (T1), the moment of venipuncture (T2), and post-IV placement (T3).

Outcomes

The primary outcome was the change in observed pain intensity between the pre- and intra-procedure stages measured using both the Face, Legs, Activity, Cry, Consolability (FLACC) scale and Evaluation Enfant Douleur (EVENDOL) scale. The FLACC and EVENDOL scales are reliable, validated behavioral scales for scoring distress in young children.[14,15] The secondary outcomes included pain scores at each time point during the needle procedure, the change in pain score between the intra- and post-procedural stages, and the satisfaction of the guardian using a five-point Likert type questionnaire after the needle procedure.

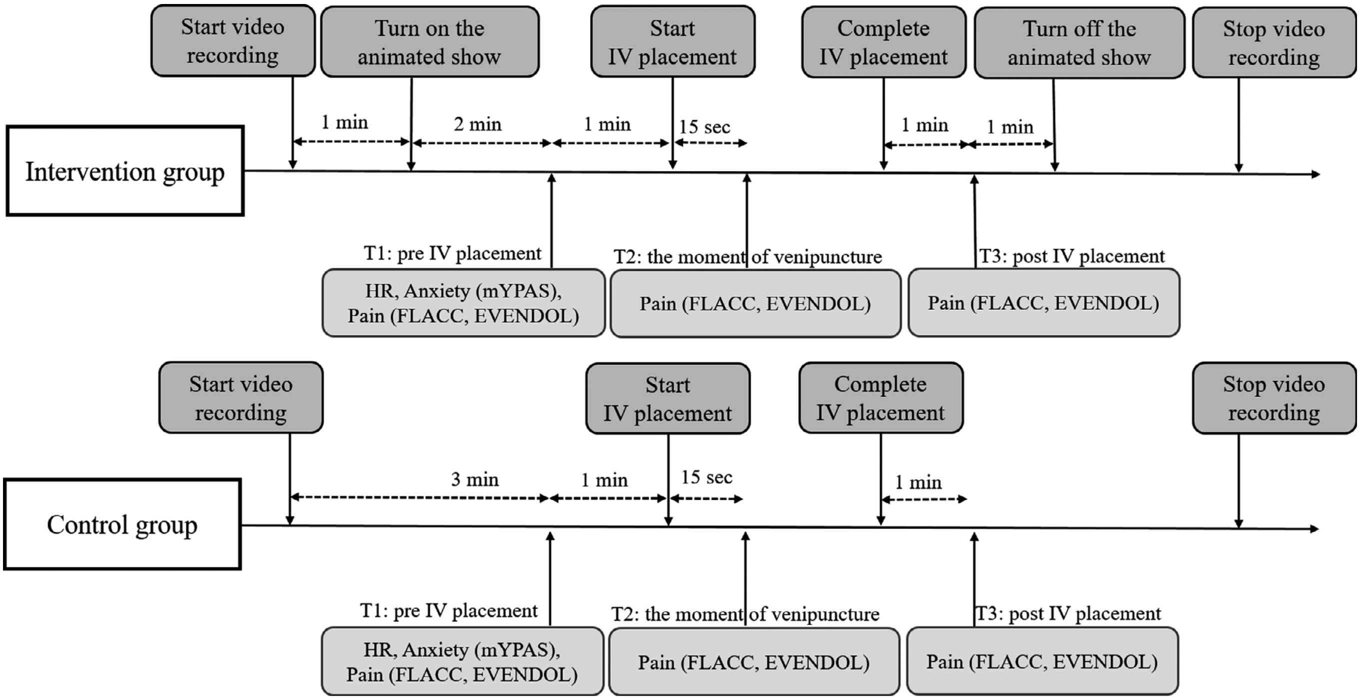

Fig. 1 Process of the study protocol. FLACC, Face, Legs, Activity, Cry, Consolidation scale; HR, heart rate; IV placement, intravenous placement; mYPAS, modified Yale preoperative anxiety scale.

Sample size

As in previous pilot studies regarding distraction methods for children during painful procedures, pre-study sample-size calculations were not performed.[16] We aimed to recruit approximately 30 children for both arms to provide sufficient preliminary evidence of the clinical efficacy of the distraction method involving tablet PC.

Statistical analysis

First, the demographic variables were described using nonparametric statistics because of the small sample size. Categorical variables are reported as frequencies and percentages, and continuous variables are reported as medians and interquartile ranges (IQRs). Second, both the FLACC scale and the EVENDOL scale scores at each time point (T1, T2, and T3) and the change scores (T2-T3, T2-T1) were compared between the intervention group and control group using the Mann–Whitney *U*-test. A between-group comparison of the guardians' responses to the post-procedural questionnaire was conducted using the Mann–Whitney *U*-test. Third, a nonparametric repeated measures analysis of variance (RMANOVA) was conducted to evaluate the effect of the distraction method using a tablet PC over the course of IV placement. All statistical analyses were two-tailed, and a P value <0.05 was considered significant. Statistical analyses were performed by using R software, version 3.6.3 (R Foundation for Statistical Computing, Vienna, Austria).

Results

According to the eligibility criteria, 214 children were eligible for inclusion, and 22 (10.3%) of these had guardians who consented to participate in this study (Fig. 2). The analysis was conducted with a total sample of 22 children (11 in the intervention group and 11 in the control group). Table 1 provides information about the demographics of the children. There were no significant differences between groups in baseline characteristics of either the guardian or children. The overall median age of the children was 4.0 (interquartile range [IQR]: 3.3–4.0) years, and 50% were girls. Approximately 91% of the children had previous experience with venipuncture.

A comparison of the FLACC scale and EVENDOL scale scores between the groups is presented in Table 2. There was no significant difference at any time during IV placement. Despite the use of a tablet PC showing an animated clip as the intervention method, the scores for both scales at all three time points were higher in the intervention group than in the control group. There were no significant differences between groups except that the degree of pain relief after the procedure measured using both the EVENDOL scale and FLACC scale was greater in the intervention group. The EVENDOL scale score change between the moment of venipuncture (T2) and T3 was 6.0 (IQR: 4.2–6.8) in the intervention group and 3.0 (IQR: 2.0–3.8) in the control group (P = 0.011). The FLACC scale score change between T2 and T3 was 4.0 (IQR: 3.0–4.2) in the intervention group and 3.0 (IQR: 1.5–3.5) in the control group (P = 0.043).

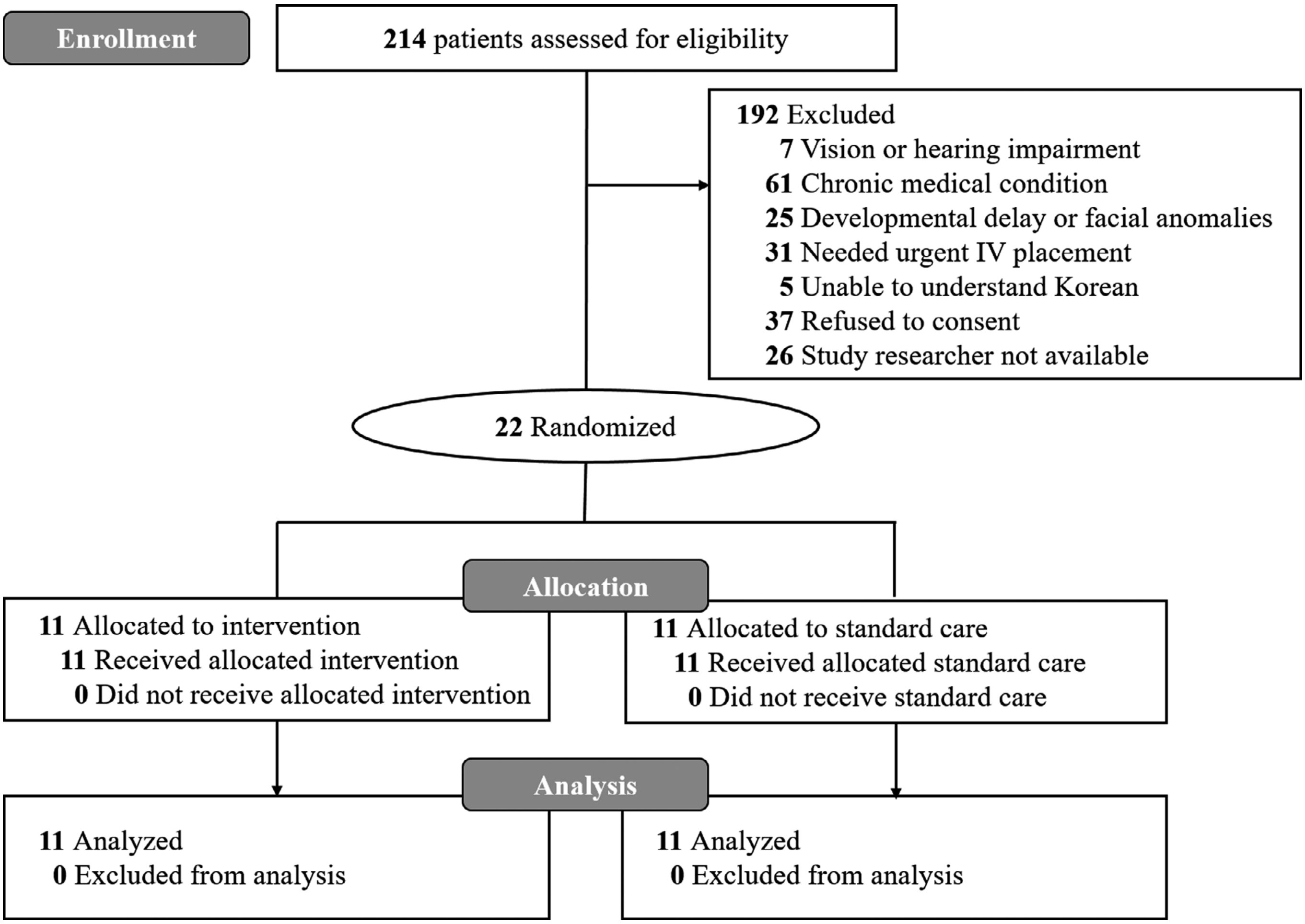

Fig. 2 Flow diagram of participants included in the analysis.

The results of the nonparametric RMANOVA showed a significant main effect of time, $F(2, 46) = 37.14$, $P < 0.001$, but a non-significant main effect of intervention, $F(1, 23) = 2.72$, $P = 0.11$. Furthermore, the time by intervention by EVENDOL score interaction was not significant, $F(2, 46) = 2.27$, $P = 0.11$, indicating that the intervention had no effect on the differences between groups over time during IV placement (Table 3). Additionally, there were also no statistically significant differences between groups in anxiety regarding the overall procedure (11.5 [IQR]: 10.2–14.8 vs. 13.0 [IQR]: 10.0–15.8, $P = 1.000$) (Table 1) or guardian satisfaction (5.0 [IQR]: 4.0–5.0 vs. 5.0 [IQR]: 3.0–5.0, $P = 0.459$) (Fig. S1).

Discussion

In this pilot study, we sought to evaluate the effectiveness of distraction using a tablet PC and animation, but there were no statistically significant differences in the pain scores between the intervention group and the control group during the procedure. Although the difference was not significant, the observed pain intensity in the intervention group tended to be higher than that in the control group at all times. Considering that the baseline heart rate was faster, and the baseline anxiety score was higher in the intervention group, the children in the intervention group were more sensitive to the fear and pain of the needle procedure than the control group. Despite an effort to achieve homogeneity of intervention (e.g., usage of same model of tablet PC and content, fixation of tablet PC position in treatment room), our results were consistent with previous study of lack of benefit for distraction during the needle procedure.[10]

Although there was no significant overall main effect of tablet PC distraction, distraction using tablet PC has shown potential to reduce children's distress after IV placement during the recovery phase. A larger change between the moment of venipuncture (T2) and the post-procedural time point (T3) in the intervention group suggested that distraction using a tablet PC may help calm the child after venipuncture. However, since the pain score at T3 was numerically higher in the intervention group, it is insufficient to conclude that tablet PC has an analgesic effect. Owing to the small number of study participants, it was impossible to show statistically meaningful results in the analysis. A larger sample size is needed to conduct the main study of PC distraction to show its effectiveness in reducing distress during IV placement.

Table 1 Demographic characteristics of study participants by group

	Control (*n* = 11)	Intervention (*n* = 11)	Total (*n* = 22)	*P* value
Age, median (IQR), years	4.0 (4.0–4.5)	4.0 (3.0–4.0)	4.0 (3.3–4.0)	0.230
Gender, *n* (%)				
Boys	7 (63.6)	4 (36.4)	11 (50.0)	0.394
Girls	4 (36.4)	7 (63.6)	11 (50.0)	
Reason for PED visit, *n* (%)				
Illness	8 (72.7)	9 (81.8)	17 (77.3)	1.000
Injury	3 (27.3)	2 (18.2)	5 (22.7)	
Previous venipuncture, *n* (%)				
No	1 (9.1)	1 (9.1)	2 (9.1)	1.000
Yes	10 (90.9)	10 (90.9)	20 (90.9)	
Baseline heart rate (HR)[†]	111.0 (104.0–116.0)	129.5 (111.0–164.0)	116.0 (104.0–133.0)	0.084
Baseline anxiety	11.5 (10.2–14.8)	13.0 (10.0–15.8)	12.3 (10.0–15.4)	1.000
Guardian's characteristics				
Age, median (IQR), years	37.0 (36.0–39.5)	37.0 (36.0–37.5)	37.0 (36.0–38.75)	0.383
Relationship, *n* (%)				
Mother	9 (81.8)	7 (63.6)	16 (72.7)	0.635
Father	2 (18.2)	4 (36.4)	6 (27.3)	
Number of children, median (IQR)	2.0 (2.0–2.5)	2.0 (1.0–2.5)	2.0 (1.0–2.8)	0.385
Experience of observation of child's venipuncture, *n* (%)				
No	2 (18.2)	3 (27.3)	5 (22.7)	1.000
Yes	9 (81.8)	8 (72.7)	17 (77.3)	

HR, heart rate; IQR, interquartile range; PED, pediatric emergency department.
[†]One child from the intervention group was excluded from the analysis due to missing data.

There were no technical issues or side effects reported during the study period. Most of the guardians in the intervention group were satisfied with tablet PC distraction, even though this was an unfamiliar service to them, which was provided in addition to medical care. A previous study by Shahid *et al.* also reported that tablet PC distraction could reduce guardians' perception of their child's distress and improve guardians' satisfaction with pain control during needle procedures.[12] However, these findings should be interpreted with caution, as these results were based on a mere impression, not an objective pain assessment tool. It would be helpful to develop and validate an objective pain assessment tool that can be used easily by guardians.

We also prepared only one animated video for the consistency of intervention but it may not have effectively drawn the interest of all children due to the varying type of interest depending on gender and age of children. Subgroup analyses (e.g., by age and gender) using specialized content for each group may suggest directions for future study. It seems that the approximately 2 min of watching time we planned was not enough for some children to focus on the animated video well. In future studies, it is necessary to give children sufficient time to concentrate fully on the animation before the start of the needle procedure. In addition, approximately 90% of children had already experienced IV placement, and some children became anxious from the moment the EMT entered the treatment room and could not be comforted by any means. Changes in the shape or location of distraction tools that can effectively block the child's field of vision from the procedure scene also need to be considered.

We could not proceed with the main trial of distraction with tablet PC because it was expected that a sufficient number of children could not be recruited. Although many guardians are already showing children's video and animated video through tablet PCs, a large number of guardians refused to participate in this study because their children had never been exposed to a tablet PC. They were reluctant to use a tablet PC and animated video as a distraction method for educational reasons. To plan further research, it seems necessary to change the methodology, for example by using content with some educational aspect rather than just recreational content or selecting older children as participants who have already been exposed to tablet PCs. Using other ways of displaying an animation (e.g., larger screens for display or augmented-reality (AR) or virtual reality (VR) environment system installed separately in the treatment room) may be alternatives to intervention to prevent children from being exposed to highly accessible tablet PC that guardians are concerned about. For younger children and infants, exploring other distraction methods that do not require advanced cognitive skills is needed, such as listening to music using headphones and applying vibrating devices (i.e., Buzzy Bee).[4]

Limitations

This study has some limitations. First, even though it was a pilot study, the sample size was too small for analysis. We could not obtain the originally planned number of participants, mainly because of the low consent rate. Second, we could not consider covariate effects on baseline pain intensity, such as the reason IV placement was needed and previous experience.

Table 2 Scores on the each observational scale of pain by group

		Group		P †
		Control (n = 11)	Intervention (n = 11)	
FLACC				
T1	Mean (SD)	2.5 (3.0)	3.6 (3.0)	0.303
	Median (IQR)	1.5 (0.0–3.8)	4.0 (0.5–5.2)	
	Range	0.0–8.5	0.0–8.5	
T2	Mean (SD)	5.3 (3.0)	7.0 (1.9)	0.186
	Median (IQR)	6.0 (3.0–7.8)	7.0 (6.8–8.5)	
	Range	0.5–9.5	2.5–9.5	
T3	Mean (SD)	2.7 (2.9)	3.3 (1.7)	0.390
	Median (IQR)	2.5 (0.2–4.8)	3.0 (2.5–4.0)	
	Range	0.0–8.0	1.0–7.0	
T2-T1	Mean (SD)	2.8 (2.0)	3.5 (2.6)	0.510
	Median (IQR)	2.5 (1.2–4.2)	3.0 (1.5–5.5)	
	Range	0.0–6.5	−0.5–7.0	
T3-T2	Mean (SD)	2.6 (1.7)	3.8 (1.3)	0.043
	Median (IQR)	3.0 (1.5–3.5)	4.0 (3.0–4.2)	
	Range	−6.0	1.5–6.0	
EVENDOL				
T1	Mean (SD)	3.5 (4.4)	4.9 (4.3)	0.390
	Median (IQR)	1.0 (0.2–5.5)	4.5 (0.8–7.0)	
	Range	0.0–14.0	0.0–12.0	
T2	Mean (SD)	7.0 (4.4)	9.6 (3.2)	0.139
	Median (IQR)	7.5 (3.5–9.8)	9.0 (8.0–12.2)	
	Range	−14.5	3.0–14.5	
T3	Mean (SD)	3.8 (4.0)	4.2 (2.8)	0.574
	Median (IQR)	3.0 (2.0–3.8)	3.5 (3.0–5.0)	
	Range	0.0–12.5	1.0–11.0	
T2-T1	Mean (SD)	3.5 (2.7)	4.8 (3.2)	0.307
	Median (IQR)	2.5 (1.5–5.5)	4.0 (2.5–8.0)	
	Range	0.5–8.0	−0.5–9.0	
T3-T2	Mean (SD)	3.2 (2.3)	5.5 (1.9)	0.011
	Median (IQR)	3.0 (2.0–3.8)	6.0 (4.2–6.8)	
	Range	1.0–9.5	2.0–8.0	

CI, confidence interval; EVENDOL, Evaluation Enfant Douleur scale; FLACC, the Face, Legs, Activity, Cry, Consolability; IQR, interquantile range; SD, standard variation.
†Mann–Whitney U-test.

Possible covariates that could affect baseline pain intensity should be considered for future studies to adequately evaluate the effectiveness of the distraction method. Third, we collected heart rate as an indicator of children's baseline pain or anxiety. However, it would also be necessary to collect information on the underlying diseases of children, the reason why IV placement was needed, other vital signs such as blood pressure, saturation, and factors that may affect baseline vital signs such as fever and degree of dehydration for an appropriate explanation of children's baseline conditions.

Table 3 Nonparametric repeated measures analysis of variance results

	Analysis of ANOVA-type statistic			
	F	df1	df2	P
FLACC				
Group	1.15	1.00	∞	0.286
Time point	35.20	1.61	∞	<0.001
Group : Time point	1.17	1.61	∞	0.302
EVENDOL				
Group	1.35	1.00	∞	0.255
Time point	34.92	1.64	∞	<0.001
Group : Time point	0.55	1.64	∞	0.542

ANOVA, analysis of variance; df, degree of freedom; EVENDOL, Evaluation Enfant Douleur scale; FLACC, Face, Legs, Activity, Cry, Consolability.

Conclusion

The results from this small pilot study showed that distraction using a tablet PC may reduce children's distress during the recovery phase after a venipuncture. However, it is difficult to use tablet PC as a distraction method in clinical practice because guardians are reluctant to expose their children to tablet PCs from an educational point of view. Further research with a larger sample size and different methods of distraction is essential.

Disclosure

The authors declare no conflict of interest.

Author contributions

S.Y. designed the study and collected data. S.Y. and H.N. performed statistical analysis. H.N. wrote the manuscript. Y.H., D.K., J.Y., and J.W. gave conceptual advice. All authors read and approved the final manuscript.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's web-site:

Fig. S1. Satisfaction scores of the guardians by the group.